

AI Workflow Audit

Purpose of the Audit

This workflow audit maps recurring study, data-analysis, reporting, and project tasks to determine where AI can be used effectively, where human judgment must remain central, and where repetitive work can be automated.

The goal is not to use AI for every task. Instead, the audit identifies the appropriate level of AI involvement based on the task's risk, repeatability, need for judgment, and verification requirements.

1. Recurring Workflow Tasks and AI Classification

The following tasks are based on my current data science, AI-fluency coursework, and content-refresh scoring project.

Recurring Task	AI Classification	Rationale
Clean and inspect datasets	Collaborate with AI	AI can suggest data-quality checks and Pandas code, while I verify the dataset and decide which issues matter.
Write Python/Pandas analysis code	Delegate to AI with review	AI can generate routine code quickly, but I must run it, inspect the output, and confirm that it matches the intended analysis.
Interpret statistical and model results	Collaborate with AI	AI can explain patterns and possible interpretations, but I must decide whether the evidence supports the conclusion.
Create distributions and analysis charts	Delegate to AI with review	AI can produce plotting code, while I check that the visualization is accurate, readable, and appropriate.
Check for data leakage	Collaborate with AI	AI can identify possible leakage risks, but I must verify feature definitions and make the final decision about what is safe to use.
Write report sections	Delegate to AI with review	AI can turn analysis notes into clear prose, but I must verify factual claims and ensure the writing represents my actual work.

Recurring Task	AI Classification	Rationale
Debug Python errors	Delegate to AI with review	AI can identify likely causes and propose fixes, but I need to run the fix and confirm that it does not introduce another problem.
Organize project files and repeated outputs	Fully automate	File creation, naming, and repeated exports are deterministic tasks that can be handled reliably by scripts.
Study difficult technical concepts	Collaborate with AI	AI can explain concepts at different levels, while I test my own understanding and verify important concepts.
Choose the final research question	Just me	The research question determines what I want to investigate and why, so the final choice requires my own academic goals and judgment.
Decide whether a result is credible enough to report	Just me	I am responsible for deciding whether the evidence is sufficient, limitations are acknowledged, and conclusions are academically defensible.
Proofread and improve assignment writing	Delegate to AI with review	AI can improve clarity, grammar, and structure, but I must ensure the final text remains accurate and reflects my own work.
Search for learning resources and documentation	Delegate to AI with review	AI can help locate relevant resources, while I verify the source and determine whether it is appropriate.
Build the ranked content-review queue	Collaborate with AI	AI can calculate scores and generate reason codes, while I validate the scoring logic and retain responsibility for the review decision.
Decide whether content should actually be changed	Just me	The final editorial or business decision requires context that the scoring dataset does not contain and should not be delegated to AI.

Key observation

The audit deliberately identifies several “**Just me**” tasks. These include choosing the research question, deciding whether evidence is credible enough to report, and deciding whether content should actually be changed. These tasks require personal judgment, accountability, context, and responsibility that should not be delegated to AI.

2. Free AI Toolkit and Course Setup

The required toolkit consists of **ChatGPT, Claude, and Anthropic Academy.**

Tool / Resource	Required Setup	Evidence
ChatGPT	Create/use a free account for analysis, coding support, explanations, and drafting.	Account/dashboard evidence if required.
Claude	Create/use a free account and configure one Claude Project.	Screenshot of configured Claude Project.
Anthropic Academy	Create/use an account and enroll in <i>AI Fluency: Framework & Foundations</i> .	Enrollment/progress screenshot.
AI Fluency course	Complete at least the first module.	Course progress/completion screenshot.

3. Claude Project Configuration

I will configure one Claude Project with custom instructions that provide stable context about my role, communication preferences, and current goals.

Claude Project Custom Instructions

WHO I AM

I am a student working on data science, machine learning, and AI-fluency projects. I regularly work with Python, Pandas, data analysis, visualization, and research reports.

TONE PREFERENCES

Be clear, concise, practical, and academically responsible. Explain technical ideas in straightforward language. Avoid unnecessary jargon and do not overstate conclusions.

CURRENT GOALS

Help me:

- complete my data science and AI-fluency coursework;
- write clear, evidence-based reports;
- debug and improve Python analysis;
- identify data-quality and leakage problems;
- interpret results without overstating what the data proves;
- build reproducible project outputs.

WORKING PRINCIPLES

When helping with analysis, distinguish observed results from interpretation and avoid claiming causation unless the evidence supports it. When providing code, make it runnable and explain important assumptions. Flag uncertainty, missing data, and possible leakage rather than silently making assumptions.

Screenshot Evidence

[INSERT SCREENSHOT OF CONFIGURED CLAUDE PROJECT HERE]

The screenshot should show the Claude Project and its custom instructions clearly enough to demonstrate that the project was configured.

4. Three Target Tasks for FL-02 Through FL-04

The following three tasks will be reused across the next phases. Each has a measurable definition of what “**done well**” means.

Target Task 1 — Data Analysis and Debugging

Task:

Use AI to support dataset inspection, Pandas/Python coding, error diagnosis, and data-quality checks.

Done well means:

The code runs without unresolved errors, the requested output is produced, data-quality and leakage checks are explicitly addressed, and I can explain the important steps and assumptions without relying on AI to interpret the result for me.

Target Task 2 — Report Writing and Interpretation

Task:

Use AI to convert analysis notes and results into clear, concise, evidence-based report sections.

Done well means:

The final section is logically structured, factual claims match the computed results, observed evidence is separated from interpretation, unsupported causal claims are removed, and the final text accurately represents my own work.

Target Task 3 — Research / Method Review

Task:

Use AI as a reviewer to identify assumptions, leakage risks, methodological weaknesses, and unsupported conclusions in the project.

Done well means:

The review identifies concrete risks or assumptions, important issues are checked against the dataset or methodology, proposed corrections are tested where applicable, and no unresolved critical leakage or unsupported-claim issues remain.

Audit Summary

This audit shows that AI is most useful in my workflow for **repetitive coding, debugging, drafting, explanation, and structured review**, provided that I check the output.

Tasks involving **research direction, credibility judgments, and final content decisions** remain with me because they require context, accountability, and judgment that should not be delegated to AI.

Fully automated work is limited to **deterministic and repeatable operations**, such as organizing files and generating standard outputs.

Submission Evidence Checklist

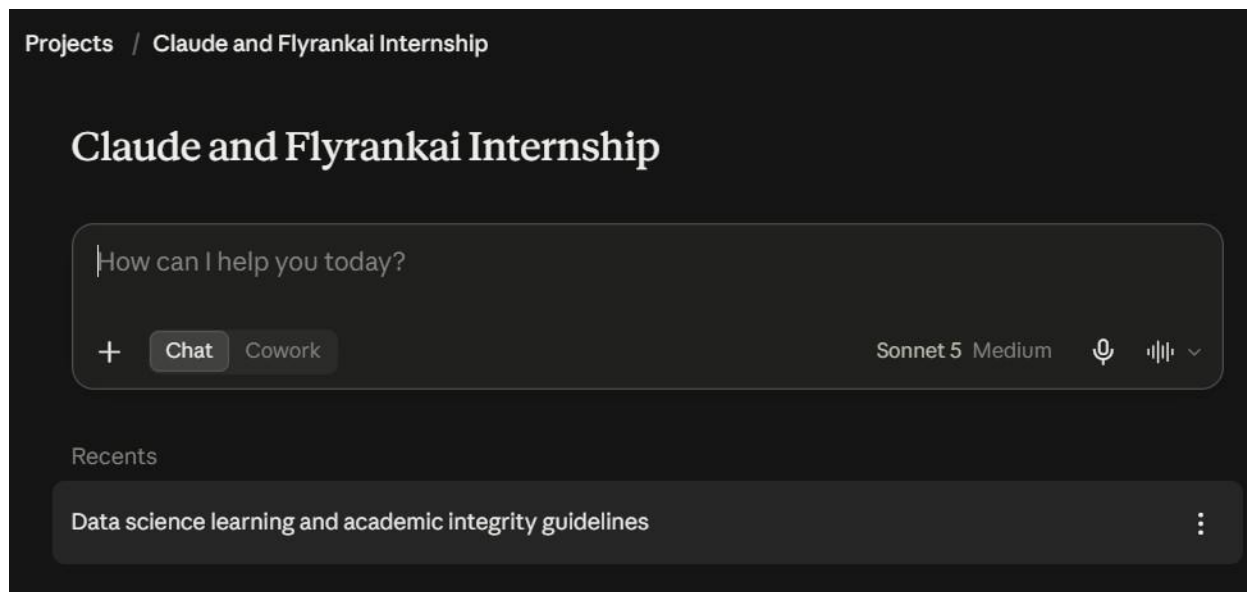
The following checklist confirms completion of the required activities and evidence for the Setup Phase.

Requirement	Status	Evidence / Notes
10+ recurring tasks genuinely related to my work/study identified	☒	15 recurring tasks documented
Every task classified as Just Me, Delegate to AI with Review, Collaborate with AI, or Fully Automate	☒	Classification and rationale provided for all 15 tasks
At least two tasks identified as “Just Me”	☒	Three tasks identified as requiring human judgment
One-line rationale provided for every task classification	☒	Rationale included in the workflow audit table

Requirement	Status	Evidence / Notes
Three target tasks selected for FL-02 through FL-04	✓	Data analysis & debugging; report writing & interpretation; research/model review
“Done Well” / success definition created for each target task	✓	Measurable success criteria documented for all three tasks
ChatGPT account/tool setup completed	✓	ChatGPT available for coursework and project work
Claude account created	✓	Claude account is active
Anthropic Academy enrollment completed	✓	Enrolled in Anthropic Academy
AI Fluency: Framework & Foundations completed	✓	Course completed
Claude Project created	✓	Project created for ongoing coursework support
Claude Project custom instructions added	✓	Project-specific working rules and instructions added
Screenshot of configured Claude Project	✓	Screenshot are inserted below

Claude Project Evidence

Screenshot 1 – Claude Project



Screenshot 2 – Claude Project Custom Instructions

Yes — here's what's active for this session:

Communication style: clear, concise, structured (headings/bullets/tables), no unsupported confidence, academic tone without unnecessary jargon.

Working rules in effect:

- No invented data, results, stats, sources, or conclusions — only what you give me or what's actually computed
- Observed results vs. interpretation vs. assumptions vs. limitations, kept separate
- Data-quality and leakage checks before any modeling
- Runnable, simple code; I flag changes to your methodology rather than silently altering it; I never claim code ran if it didn't
- No causal language unless evidence supports it; no "significant/predictive/reliable" claims unless earned
- No fabricated citations; your original meaning preserved when editing
- In review mode: I check for leakage, unsupported claims, calculation errors, methodology weaknesses, missing limitations, reproducibility issues — and explain what's wrong → why it matters → how to fix it
- Human judgment stays central — I flag what needs your verification rather than presenting outputs as decisions

Dataset-specific rules noted (rate columns already $\times 100$, `avg_position=0` = missing, `scroll_rate` / `ai_traffic_pct` can exceed 100 legitimately, `trend_direction` / `trend_pct` are leakage relative to `is_declining_label`, missingness tied to `content_type` so no blind `fillna(0)`, IDs are grouping-only never features).

One caveat worth being upfront about: this is all held in my context for **this conversation only** — I don't retain it automatically across sessions unless you have Claude's memory feature enabled in settings.

Ready when you are — send the dataset file and/or the assignment brief whenever you have them.

Figure 2. Claude Project configuration and custom instructions — submitted as screenshot evidence for the Setup Phase.

Completion Statement

All required Setup Phase workflow-audit activities have been completed. The remaining submission evidence consists of the screenshots documenting the configured Claude Project and its custom instructions.

Linked Resources

- **On-boarding your AI Intern — Ethan Mollick:**
<https://www.oneusefulthing.org/p/on-boarding-your-ai-intern>
- **AI Fluency: Framework & Foundations — Anthropic Academy:**
<https://anthropic.skilljar.com/ai-fluency-framework-foundations>
- **What are Projects? — Claude Help Center:**
<https://support.claude.com/en/articles/9517075-what-are-projects>